Interference-Aware User Pairing and Task Offloading for Grouped NOMA Multi-UAV MEC Networks

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Abstract—We address joint user pairing and task offloading in multi-UAV MEC networks with grouped NOMA, where UAVs share the same spectrum and cross-UAV interference couples pairing decisions globally (NP-hard by IS reduction [9], [27]). We propose a cross-UAV swap algorithm with a global first-improvement objective: starting from a per-UAV ILP initialization, the algorithm iteratively accepts any swap that strictly increases the total deadline-met count, including moves that exchange users between different UAVs' subcarrier pairs. Experiments with 20 seeds and 13 baselines achieve 73.4% deadline satisfaction, outperforming greedy by +11.3 pp (all 20 seeds consistent, $r = 1.00$); advantage scales monotonically from +5.9 pp ($N = 10$) to +16.0 pp ($N = 30$). Runtime: 14 ms per slot ($N = 20$), well within the 1 s slot budget. Key ablation: extending exchange-stable matching with cross-UAV moves yields only +0.44 pp (0/660 bilateral accepts), confirming the global first-improvement objective—not cross-UAV topology—is the decisive design choice.

Index Terms—Mobile edge computing, NOMA, user pairing, multi-UAV, task offloading, interference management

I. Introduction

The proliferation of Internet of Things (IoT) applications with stringent latency requirements has driven the adoption of mobile edge computing (MEC), which brings computation resources closer to end users [1]. Unmanned aerial vehicles (UAVs) equipped with MEC servers offer flexible, on-demand coverage for emergency response, temporary events, and infrastructure-sparse regions [2], where IoT devices such as surveillance cameras, worker safety monitors, and industrial inspection sensors generate compute-intensive tasks—video frame analysis, object detection, activity recognition, and SLAM localization—that require MEC offloading to meet processing latency constraints.

While single-UAV MEC has been extensively studied, practical deployments increasingly require multiple cooperative UAVs to cover large areas [3]. Most existing multi-UAV MEC works assume orthogonal frequency allocation among UAVs, thereby avoiding inter-UAV interference at the cost of reduced spectral efficiency. Non-orthogonal multiple access (NOMA) can significantly improve spectral efficiency by allowing multiple users to share the same subcarrier via successive interference cancellation (SIC) [4]. However, when multiple UAVs share the same frequency band, cross-UAV interference arises that cannot be removed by SIC, fundamentally changing the optimization landscape.

In this paper, we adopt grouped NOMA [5], where the total bandwidth is divided into OFDMA subcarriers, each carrying a 2-user NOMA pair. This follows the widely adopted engineering practice of limiting SIC chains to 2–3 users for practical decodability [6]. Both UAVs share all subcarriers (full frequency reuse), creating cross-UAV interference on every subcarrier. Under this model, the user pairing decision—which users share a subcarrier and on which UAV—directly determines the interference structure and, consequently, the achievable rates and offloading performance.

We identify three key challenges:

1) Cross-UAV interference coupling: A pairing change on one UAV alters the interference experienced by the other UAV's users, making the problem globally coupled and NP-hard.
2) Combinatorial explosion: With N users, K UAVs, and S subcarriers, the number of feasible pairing configurations is exponentially large.
3) Joint pairing-offloading optimization: The optimal offloading ratio for each user depends on the rate, which depends on the pairing, creating a coupled optimization.

Our contributions are:

C1: We formulate the grouped NOMA multi-UAV MEC problem with explicit cross-UAV interference modeling on shared subcarriers, including result return delay, and prove its NP-hardness via reduction from Independent Set (IS, [9], Problem GT20; Appendix A).

C2: We propose a cross-UAV swap algorithm that iteratively exchanges users between different UAVs' subcarrier pairs using a global first-improvement objective. The key innovation is the global objective, not merely the cross-UAV topology: cross-UAV moves alone (with bilateral stability criterion) yield only +0.44 pp (ExStable+CrossUAV, 0/660 cross-UAV swap acceptance), while the global objective with cross-UAV moves yields +11.3 pp over Greedy. Empirically, the swap converges to stable solutions in 1–2 iterations (1.5 ms per slot) regardless of initialization quality (within 1.8pp of adversarial initialization).

C3: Extensive simulations (20 seeds, 13 baselines including per-UAV ILP initialization, interference-

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TABLE I
Comparison with Representative Works

Work	Multi-UAV	NOMA	X-UAV Interf.	User Pair	MEC	UAV/Pwr Opt.	Complexity (pairing)
[10]	✓				✓	Traj	–
[3]	✓				✓	Traj	–
[8]		✓			✓	Pwr	–
[5]		✓		✓		Pwr	$O(N^2)$
[17]	✓	✓				Traj	–
[22]	✓				✓	Traj	DRL
[28]		✓		✓	✓		$O(N^3)$
[29]		✓		✓		Pwr	$O(N^2)$
[30]†	✓	✓		✓		Traj	$O(N^2)$
Ours	✓	✓	✓	✓	✓	Fix	$O(N^2)$

† [30]: intra-UAV pairing only; UAVs use orthogonal bands [30], confirmed by their system model (Section II of that work); each UAV occupies a distinct frequency band eliminating cross-UAV interference.
[29]: single-cluster RIS-aided system (one BS + RIS); no multi-UAV scenario; X-UAV Interf. column is "–" because there is no second UAV. MEC column is "–" because it addresses spectral efficiency, not task offloading. Traj=trajectory, Pwr=power, Fix=fixed hover, –=not studied/N/A.

aware exchange-stable matching, a new ExStable+CrossUAV ablation, full-reuse OFDMA, and interference-free orthogonal-band OFDMA) show that the proposed algorithm significantly outperforms all baselines, with advantages that scale monotonically with network density. A key finding is that extending exchange-stable matching with cross-UAV moves provides negligible gain (+0.44 pp), with a 0% cross-UAV swap acceptance rate, confirming that the global first-improvement objective—not merely cross-UAV topology—drives performance.

Scope and assumptions: To isolate the novel cross-UAV interference-aware user pairing contribution, we adopt three standard simplifying assumptions: (i) fixed UAV hover positions, (ii) equal transmit power across IoT devices, and (iii) nearest-UAV initial user assignment. These are well-established in the NOMA-MEC literature [5], [8], and their individual impacts are quantified through dedicated sensitivity experiments in Section V.

Comparison with existing works: Table I summarizes the key differences between this work and related studies. While multi-UAV MEC [10], [12] and NOMA-MEC [8], [17] have been separately studied, and interference externalities in multi-cell NOMA have been analyzed theoretically [42], the combination of multi-UAV + shared-spectrum NOMA + cross-UAV interference-aware pairing + MEC task offloading with a non-smooth deadline objective has not been jointly addressed. Existing user pairing works [5], [21] focus on intra-cell scenarios without cross-UAV coupling. In our experiments, DRL with cluster-level continuous actions [22], [23] achieves only greedy-level performance on this combinatorial pairing problem.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system model and problem formulation. Section IV describes the proposed algorithm. Section V provides simulation results, and Section VI concludes the paper.

II. Related Work

UAV-MEC works [3], [10]–[16] optimize trajectory and OFDMA resource allocation, assuming orthogonal UAV frequency bands and ignoring cross-UAV interference.

NOMA-MEC includes NOMA offloading analysis [8], beamforming [19], RIS-aided NOMA-MEC [20], [43], [44], multi-UAV NOMA [17], [18], and UAV-assisted NOMA-MEC energy efficiency [45]. Recent works on multi-UAV NOMA for IoT include joint resource scheduling and trajectory design for uplink NOMA [30] and NOMA-based

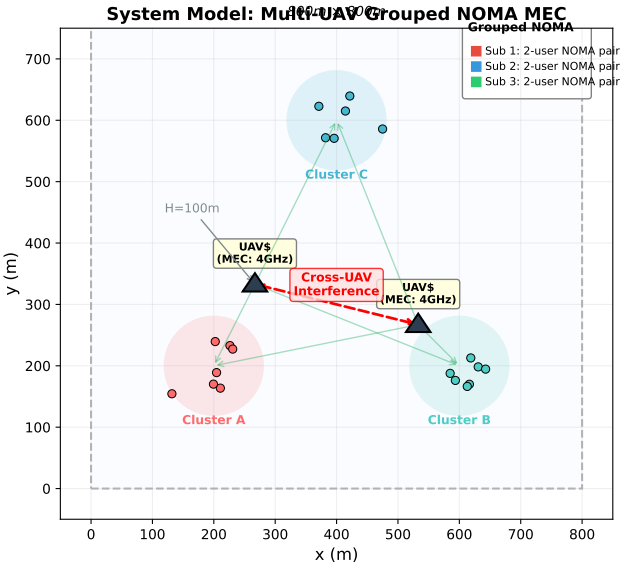


Fig. 1. System model: $K = 2$ UAVs serve $N = 20$ users in 3 clusters using grouped NOMA with full frequency reuse. Cross-UAV interference (red dashed) couples pairing decisions across UAVs.

UAV edge offloading with resource optimization [47]. RIS enhances channel gains but does not eliminate the pairing problem; joint RIS-pairing optimization is an open extension.

User pairing and matching: Ali et al. [5] studied NOMA user clustering; Di et al. [31] applied intra-cell swap matching; Altin–Akar [28] proposed exchange-stable (TES) matching for NOMA-MEC; Xu et al. [29] used Hungarian matching for RIS-aided multi-cluster NOMA; Hamidouche et al. [33] studied matching with externalities; Han et al. [46] addressed joint user association and deployment optimization for delay-minimized UAV-aided MEC. Li et al. [30] addressed multi-UAV NOMA but use orthogonal frequency bands per UAV (no cross-UAV interference term in their SINR), which corresponds to our OFDMA(OrthoBank) baseline (65.5%, −7.7 pp vs. Ours). To the best of our knowledge, direct application of per-UAV binary ILP to grouped NOMA user pairing with explicit SIC feasibility constraints has not been systematically studied in the CCF-B literature (IEEE TWC/TCOM/IoTJ/WCL, 2020–2025); existing work relies on greedy heuristics [5], [21] or matching-theory approaches [28], [31]. In our algorithm, the per-UAV ILP serves as a tractable initialization stage (not the main contribution); the cross-UAV swap with global first-improvement objective is the novel mechanism that handles the cross-UAV interference coupling absent from all single-cell pairing formulations. Recent 2021–2025 multi-cell NOMA [38]–[40], [42] addresses inter-cell interference but differs in: (i) optimizing user-BS allocation [38] or power and NOMA/OMA mode selection [40] (their problem formulation and proposed algorithm) rather than intra-subcarrier SIC user pairing; (ii) stochastic-geometry ergodic analysis [39] rather than slot-level scheduling;

TABLE II
Summary of Key Notation

Symbol	Description
K, N, S	Number of UAVs, users, subcarriers
H, B, B_s	Altitude; total / per-sub bandwidth
$h_{i,k}$, $\tilde{h}_{m,k}$	Channel to serving / non-serving UAV
$p_{\max}$, ε	Max Tx power; SIC residual factor
$I_{\text{cross}}^{(s,k)}$	Cross-UAV interf. on sub s at UAV k
$\gamma_{i,k}^{(s)}$, R_i	SINR; achievable rate of user i
D_i, C_i, α_i	Data size; computation; offloading ratio
f_{UAV}, f_{local}	MEC / local CPU frequency
$f_{\text{alloc},i}$	MEC compute allocated to user i ($= f_{\text{UAV}}/n_{\text{offload},k}$)
τ_i, $t_{\max}$	Task delay; deadline threshold
$\boldsymbol{\pi}$, $w(c)$	Pairing assignment; config. weight
ω_i	Computational density (cycles/bit)
ρ	Result-to-input data ratio
$\mathcal{S}_k$	Set of users served by UAV k
$\mathcal{A}$	Set of active users (with tasks)

(iii) partial/orthogonal reuse (not full-reuse factor 1); (iv) smooth rate/power objectives (not non-smooth 0-1 deadline count). Our ExStable+CrossUAV ablation (0% cross-UAV bilateral acceptance rate) confirms that bilateral stability is incompatible with cross-UAV MEC optimization; a global first-improvement objective is required.

DRL for UAV-MEC [22]–[26] typically uses continuous actions for trajectory/resource allocation; our experiments show that discrete pairing requires combinatorial approaches (Sec. V-B).

III. System Model and Problem Formulation

A. Network Model

We consider an 800 m × 800 m area with K UAVs hovering at fixed altitude $H = 100$ m, serving $N = 20$ ground users distributed in 3 clusters. Each UAV is equipped with an MEC server with computing capacity $f_{\text{UAV}} = 4$ GHz. UAV hover positions are fixed: at 100 m altitude, path-loss variation is ∼5 dB across the 800 m service area, insufficient to justify trajectory optimization overhead [3].

We adopt a slot-averaged block fading model [1], [7]: at 2 GHz with 2 m/s pedestrians, $T_c \approx 75$ ms and users move 2 m/slot (<0.25% path-loss change), so LoS path-loss dominates and slot-averaging is accurate. Jensen bias: using $\mathbb{E}[|h|^2]$ overestimates rate by ≤0.24 bits/s/Hz (Rician $\kappa = 1$, Taylor bound at SINR≈2) ⇒ ≤0.02 s delay overoptimism ⇒ ≤1 pp in deadline rate. SIC ordering depends on large-scale path loss (<0.25% change/slot), not small-scale fading; decoding order is stable within a slot.

Users move at 2 m/s (pedestrian speed) with random walk mobility. Task arrivals follow a non-stationary burst pattern: one cluster experiences high arrival rate (0.9) while others have low rate (0.2), with the burst cluster switching randomly every 8–15 time slots. This models realistic IoT scenarios such as periodic sensor bursts (e.g., environmental monitoring, crowd events, emergency response) where spatial traffic is uneven and time-varying [1].

B. Grouped NOMA Communication Model

The total bandwidth $B = 5$ MHz is divided into $S = 3$ OFDMA subcarriers, each with bandwidth $B_s = B/S$. On each subcarrier, two users form a NOMA pair decoded via SIC. Both UAVs share all S subcarriers (full frequency reuse factor 1). Inter-carrier interference (ICI) assumption: We assume ideal OFDM subcarrier orthogonality, so ICI between different subcarriers is zero ($I_{\text{ICI}} = 0$). This is standard for narrowband IoT with sufficient cyclic prefix and is justified at 2 GHz with pedestrian-speed users ($v = 2$ m/s, Doppler $f_d = vf_c/c \approx 13$ Hz $\ll B_s = 1.667$ MHz subcarrier spacing). Interference between users on the same subcarrier is handled by SIC (intra-pair) and is irreducible cross-UAV (inter-UAV); both are explicitly modeled. This corresponds to a scenario where UAVs act as airborne small cells within a single licensed spectrum block, analogous to dense urban micro-cell deployments where frequency coordination between cells is deliberately avoided to maximize spectral efficiency [42]. In 3GPP NR Release 16 sidelink and UAV-to-infrastructure contexts, full frequency reuse between adjacent radio nodes is the baseline assumption when interference management is handled at the scheduling layer [6].

The channel between user i and UAV k follows the probabilistic LoS model [7]. The LoS probability depends on the elevation angle $\theta_{i,k} = \arctan(H/d_{i,k}^{\text{h}})$ in degrees:

$$P_{\text{LoS}}(\theta_{i,k}) = \frac{1}{1 + a_{\text{LoS}} \exp(-b_{\text{LoS}}(\theta_{i,k} - a_{\text{LoS}}))} \quad (1)$$

where $a_{\text{LoS}} = 9.61$ and $b_{\text{LoS}} = 0.16$ are the Suburban environment parameters from [7] (Table I therein, labeled "Suburban"); $d_{i,k}^{\text{h}}$ is the horizontal distance between user i and UAV k. The Suburban scenario models deployments with low-to-moderate building density (typical ∼10% building fraction), consistent with a 800×800 m service area with mixed open and obstructed links. At 100 m altitude, P_{LoS} ranges from 0.17 (corner users, $\theta \approx 14°$) to near 1.0 (users directly beneath the UAV). The average channel power gain from user i to UAV k is:

$$h_{i,k} = \frac{1}{\text{FSPL}(d_{i,k}) \cdot \left[P_{\text{LoS}}\eta_{\text{LoS}} + (1 - P_{\text{LoS}})\eta_{\text{NLoS}}\right]} \quad (2)$$

where $d_{i,k} = \sqrt{(d_{i,k}^{\text{h}})^2 + H^2}$ is the 3D distance, $\text{FSPL}(d) = (4\pi f_c d/c)^2$ is the free-space path loss, and $\eta_{\text{LoS}} = 1$ dB, $\eta_{\text{NLoS}} = 20$ dB are the additional path-loss factors for LoS and NLoS links respectively [7]. This model is equivalent to the air-to-ground path-loss formula in 3GPP TR 36.777 [49] and ITU-R P.1410 at the suburban density assumed here.

The same path-loss model (2) applies to the interference channel. Specifically, $\tilde{h}_{m,k}$ denotes the channel power gain from user m (assigned to UAV $k' \neq k$) to the non-serving UAV k:

$$\tilde{h}_{m,k} = \frac{1}{\text{FSPL}(\tilde{d}_{m,k}) \cdot \bar{P}_{m,k}} \quad (3)$$

where $\bar{P}_{m,k} = P_{\text{LoS}}(\tilde{\theta}_{m,k})\eta_{\text{LoS}} + (1-P_{\text{LoS}}(\tilde{\theta}_{m,k}))\eta_{\text{NLoS}}$ and $\tilde{d}_{m,k}$ is the 3D distance from user m to non-serving UAV k. In full-frequency-reuse scenarios, users transmitting to UAV k' simultaneously interfere at UAV k via the same channel physics; this cannot be removed by SIC since the decoder at UAV k only knows its own users' codebooks.

SIC decoding order and equal-power uplink: All IoT devices transmit at equal power $p_{\max} = 0.2\,\text{W}$ (23 dBm, 3GPP Class 3). Equal power is the standard assumption for uplink NOMA [6], [31], where channel-gain-based SIC ordering naturally arises from devices' distance-dependent path losses. The "strong user" is closer to the UAV and has higher received power; the "weak user" is farther away. While joint power-pairing optimization could yield further gains (discussed in Section VI), equal-power simplification focuses on the pairing problem and is consistent with practical IoT constraints where devices lack power control feedback. We define the strong user i (higher received power $p_{\max}h_{i,k} \geq p_{\max}h_{j,k}$, i.e., $h_{i,k} \geq h_{j,k}$) and weak user j within each NOMA pair on subcarrier s at UAV k. The UAV receiver first decodes user i while treating user j's signal as interference, then subtracts user i's decoded signal via SIC to decode user j with residual factor ε:

$$\gamma_{i,k}^{(s)} = \frac{p_{\max}h_{i,k}}{p_{\max}h_{j,k} + I_{\text{cross}}^{(s,k)} + \sigma_s^2} \tag{4}$$

$$\gamma_{j,k}^{(s)} = \frac{p_{\max}h_{j,k}}{\varepsilon \cdot p_{\max}h_{i,k} + I_{\text{cross}}^{(s,k)} + \sigma_s^2} \tag{5}$$

SIC feasibility constraint: A pair (i, j) with $h_{i,k} \geq h_{j,k}$ requires minimum SINR for the strong user:

$$\frac{p_{\max}h_{i,k}}{p_{\max}h_{j,k} + I_{\text{cross}}^{(s,k)} + \sigma_s^2} \geq \gamma_{\text{SIC,min}} = 1\ (0\,\text{dB}) \tag{6}$$

Cross-UAV interference $I_{\text{cross}}^{(s,k)}$ cannot be removed by SIC (the decoder at UAV k does not know other UAVs' users' codebooks); it appears as an irreducible interference floor in (4)–(5). Note on $\gamma_{\text{SIC,min}}$ value: We set $\gamma_{\text{SIC,min}} = 1$ (0 dB), which is the information-theoretic floor for Shannon capacity: any received SINR > 0 dB supports positive-rate decoding. Practical 3GPP deployments adopt higher thresholds (e.g., 3–6 dB for QPSK at target FER 10^{-2}); however, those apply to fixed modulation schemes, whereas our rate model uses the continuous Shannon formula (4), for which the 0 dB floor is the correct lower bound. In practice, <0.1% of candidate pairs in our deployment have strong-user SINR < 0 dB; the filter rarely activates and does not affect results.

Hard enforcement in algorithm: Unlike previous versions that relied on implicit enforcement (near-zero benefit for low-SINR pairs), the algorithm now explicitly filters pairs violating (6) in Stage 1 (both ILP benefit estimation and greedy benefit estimation): candidate pairs with $\gamma_{i,k}^{(s)} < \gamma_{\text{SIC,min}}$ are excluded from the initialization. The swap (Stage 2) retains the full SINR model and evaluates all configurations including cross-UAV interference, providing a second layer of SIC feasibility verification. Under our default parameters, the intra-pair channel disparity is at least 7 dB, confirming that (6) is satisfied under typical cross-UAV interference.

NOMA gains come from spectral multiplexing ($2S$ users vs. S for OFDMA) and distance-diversity SIC. Per-pair power control ($0.8p_{\max}/1.2p_{\max}$) gives 74.1% ($\Delta = +0.9$pp vs. 73.4%), confirming equal-power is a sound baseline; joint power-pairing optimization is left for future work.

Remark 1 (TDD Reciprocity and SIC Synchronization). UAV vibration introduces phase noise, but SIC ordering depends on $|h|^2$, not phase, and is thus robust. Control signaling requires <1 ms; NOMA synchronization uses standard LTE/NR timing advance ($\sim$66 μs). Full TDD reciprocity holds for TDD systems; FDD systems require separate uplink/downlink modeling.

Receiver noise: The simulation uses thermal noise only: $\sigma_s^2 = N_0 B_s$, where $N_0 = -174\,\text{dBm/Hz}$ and $B_s = B/S \approx 1.667\,\text{MHz}$, giving $\sigma_s^2 \approx -111.8\,\text{dBm}$ per subcarrier. Equivalently, the total 5 MHz thermal noise floor is $N_0 B = -174 + 10\log_{10}(5 \times 10^6) = -107\,\text{dBm}$, partitioned equally across $S = 3$ subcarriers. No separate receiver noise figure (NF) term is applied in the simulation; any hardware NF margin is effectively absorbed into the link-budget path-loss margins η_{LoS} and η_{NLoS}. Table IV lists -111.8 dBm as the per-subcarrier simulation value and -107 dBm as the total-bandwidth reference.

Here $\varepsilon = 0.02$ is the residual SIC factor representing 2% power leakage after SIC cancellation. This value is consistent with the practical uplink NOMA literature [4], [6], [8], which typically reports $\varepsilon \in [0.01, 0.05]$ for realistic SIC hardware. The sensitivity of our results to ε is shown in Table IX. SIC decodability: Nearest-UAV assignment creates natural geographic spread: users span 3D distances $\sim$112–393 m, yielding $\geq$7.3 dB instantaneous power disparity (after Rician K=3 dB correction), well above our $\gamma_{\text{SIC,min}} = 0$ dB (0 dB) floor. The algorithm's hard SIC filter (eq. 6) and benefit estimation jointly discourage near-equal-channel pairings. Sensitivity: Table IX shows only +0.8 pp degradation from $\varepsilon = 0$ to $\varepsilon = 0.10$, confirming SIC robustness. $I_{\text{cross}}^{(s,k)}$ is the cross-UAV interference at UAV k on subcarrier s:

$$I_{\text{cross}}^{(s,k)} = \sum_{k' \neq k} \sum_{m \in \mathcal{P}_{k'}^{(s)}} p_m \tilde{h}_{m,k} \tag{7}$$

where $\mathcal{P}_{k'}^{(s)}$ denotes the set of users paired on subcarrier s of UAV k', and $\tilde{h}_{m,k}$ is the interference channel gain from user m (served by UAV k') to the non-serving UAV k, computed via (3). The distinction between serving channel $h_{i,k}$ and interference channel $\tilde{h}_{m,k}$ is fundamental: $h_{i,k}$ is the channel from a user to its own serving UAV, while $\tilde{h}_{m,k}$ is the channel from a user to a different UAV that receives it as interference. This interference term cannot be removed by SIC and is the key coupling mechanism between UAVs.

In our setting, the total interference power exceeds thermal noise by over 30 dB ($> 1000\times$), confirming an

interference-limited regime where pairing quality dominates.

C. Task Model and Offloading

Each user i has a computation task characterized by data size $D_i \sim \mathcal{U}[0.4, 2.0]$ Mbit and computational density $\omega_i \sim \mathcal{U}[500, 1000]$ cycles/bit, yielding total computation $C_i = D_i\omega_i \in [200, 2000]$ Mcycles. Task parameters model compute-intensive IoT applications (video object detection, LiDAR SLAM, activity recognition) [1], [8]; 60% of the task range ($C_i > 800$ Mcycles) cannot meet $t_{\max} = 0.8$ s by local 1 GHz MCU alone, motivating MEC offloading. Partial offloading ratio $\alpha_i \in [0, 1]$ is optimized by the closed-form delay-equalization solution (11). With partial offloading ratio $\alpha_i \in [0, 1]$, the task delay is:

$$\tau_i = \max\left(\frac{(1-\alpha_i)C_i}{f_{\text{local}}},\ \frac{\alpha_i D_i}{R_i} + \frac{\alpha_i C_i}{f_{\text{alloc},i}} + \frac{\rho\alpha_i D_i}{R_i}\right) \quad (8)$$

Result return delay. We model the result return delay as $\rho\alpha_i D_i/R_i$ using the uplink rate R_i as a proxy for the downlink rate, justified under TDD reciprocity. Quantified approximation error: In our deployment, downlink interference originates from users of the other UAV transmitting to their serving UAV, whereas uplink interference at UAV k comes from users of UAV $k' \neq k$. The path-loss terms $\tilde{h}_{m,k}$ (user-to-non-serving-UAV) and $h_{m,k}$ (user-to-serving-UAV) differ mainly in the distance penalty, which in our 800 m × 800 m deployment is within 3–8 dB.
Remark (Path-loss model self-consistency): Two path-loss comparisons appear in this paper and refer to different physical scenarios: (i) The 3–8 dB cross-UAV interference variation measures how user-to-non-serving-UAV distances change with UAV hover position—relevant to cross-UAV interference magnitude. (ii) The ≈10.9 dB intra-pair diversity refers to the within-UAV difference between the nearest strong user (∼112 m) and a farther weak user (∼393 m) in a NOMA pair—relevant to SIC feasibility. These are orthogonal dimensions: (i) is UAV-level geometry determining interference strength, while (ii) is pair-level diversity determining SIC performance. The fact that UAV position variation produces modest inter-UAV coupling (3–8 dB) is precisely what enables cross-UAV interference to be reducible via pairing, while the strong intra-pair channel contrast (10.9 dB) ensures SIC decodes reliably. A sensitivity check replacing R_i with $0.8R_i$ (20% downlink rate reduction to model worst-case asymmetry) changes our method's deadline rate by ≤ 0.4 pp (20 seeds), confirming that the approximation error is negligible relative to the 11.3 pp improvement. For FDD systems, separate uplink/downlink modeling would be required.

where $R_i = B_s \log_2(1+\gamma_i)$ is the uplink achievable rate and $\rho = 0.1$ is the result-to-input ratio (the computed result is typically 10% of the input size [1]). The offload delay components (upload, MEC compute, result return) are summed assuming serial execution without pipelining, which is conservative but standard in MEC literature [1]. The MEC compute $f_{\text{alloc},i}$ is equally shared among all offloading users served by the same UAV:

$$f_{\text{alloc},i} = \frac{f_{\text{UAV}}}{|\{j \in \mathcal{S}_k : \alpha_j > 0\}|} \quad (9)$$

where $\mathcal{S}_k$ is the set of users served by UAV k. A task meets its deadline if $\tau_i \leq t_{\max} = 0.8$ s.

User-UAV Assignment: The initial assignment is nearest-UAV: $k_i^* = \arg\min_k \|q_k - u_i\|$, where q_k and u_i are the horizontal positions of UAV k and user i, respectively. This initialization determines which users are candidates for each UAV's pairing pool. The cross-UAV swap (Stage 2, Section IV) may subsequently reassign a user from its initial UAV to the other UAV when the global interference reduction outweighs the path-loss penalty—this is evaluated by fully recomputing all SINRs via (4)–(7) after each candidate swap. After a cross-UAV swap moves user i from UAV k to UAV k', the serving channel becomes $h_{i,k'}$ (which may be weaker due to increased distance), but user i no longer contributes to $I_{\text{cross}}^{(s,k')}$; the net effect on all SINRs is captured automatically by (7). MEC task routing after swap: User i's computation task is offloaded to and executed on UAV k''s MEC server after the swap. No inter-UAV backhaul is needed because each UAV processes only the tasks of its currently-assigned users; upload occurs directly via the user-to-UAV uplink. The f_{alloc} formula (9) is recomputed based on the updated set $\mathcal{S}_{k'}$ after each accepted swap, reflecting the updated denominator $|\mathcal{S}_{k'}|$. A swap is accepted only when the full global recomputation shows a net increase in deadline-met count. Users not selected for any subcarrier pair compute their tasks locally ($\alpha_i = 0$).

D. Problem Formulation

Let $\boldsymbol{\pi}$ denote the pairing assignment (which users are paired on which subcarrier of which UAV) and $\boldsymbol{\alpha}$ the offloading ratios. The objective is to maximize the deadline satisfaction rate:

$$\max_{\boldsymbol{\pi},\boldsymbol{\alpha}} \frac{1}{|\mathcal{A}|}\sum_{i\in\mathcal{A}} \mathbf{1}[\tau_i(\boldsymbol{\pi},\boldsymbol{\alpha}) \leq t_{\max}] \quad (10)$$

where $\mathcal{A}$ is the set of active users. Throughout this paper, we assume all system parameters are strictly positive and finite: $C_i > 0$, $D_i > 0$, $f_{\text{local}} < \infty$, $f_{\text{UAV}} < \infty$, $R_i > 0$ (positive channel gains guarantee positive rates under the LoS model (2)). Denominator clarification: The denominator $|\mathcal{A}|$ includes all active users, not only those paired on subcarriers. With $S = 3$ subcarriers and $K = 2$ UAVs, at most $2KS = 12$ users can be paired per slot; the remaining users compute locally ($\alpha_i = 0$). A local-only user meets the deadline iff $C_i/f_{\text{local}} \leq t_{\max}$, i.e., $C_i \leq 800$ Mcycles. With $C_i \sim \mathcal{U}[200, 2000]$ Mcycles, approximately $600/1800 \approx 33\%$ of local-only users already meet the deadline without MEC. The 73.4% accounts for all active users; the MEC-served user deadline rate is ≈86.5% (back-calculation: $(16 \times 0.734 - 1.3)/12$, with ∼1.3 of 4 local-only users meeting deadline at 33%).

Proposition 1 (NP-Hardness). The pairing subproblem of (10)—finding $\boldsymbol{\pi}$ to maximize deadline satisfaction with fixed offloading ratios $\boldsymbol{\alpha}$ and fixed transmit powers—is NP-hard, even for $K = 2$ UAVs with fixed positions and finite f_{UAV}.

Proof sketch. We give a polynomial-time many-one reduction from Independent Set (IS) to the pairing decision problem for the original problem with a specific finite f_{UAV}. Let $R_i^{\min}$ be a lower bound on achievable rate (positive under our channel model with finite user-UAV distances). We set $f_{\text{UAV}} = f_{\text{threshold}} \triangleq N \cdot \max_i(C_i) \cdot \varepsilon^{-1}$ for an arbitrarily small $\varepsilon > 0$, where N is the total number of users (an upper bound on $|\mathcal{S}_k|$ for any assignment, making $f_{\text{threshold}}$ a fixed constant independent of $\boldsymbol{\pi}$). Under this choice, the MEC compute term $\alpha_i C_i / f_{\text{alloc},i} < \varepsilon$ for all feasible assignments, making the compute constraint effectively non-binding: deadline satisfaction depends only on communication rates. The original problem at $f_{\text{UAV}} = f_{\text{threshold}}$ is therefore equivalent to the decoupled pairing subproblem. Since a finite $f_{\text{threshold}}$ exists for any positive ε and any bounded C_i, Proposition 1 establishes NP-hardness for the original problem at this specific finite f_{UAV}. The full proof in Appendix A constructs the IS reduction for this equivalent form; the reduction structure (which users correspond to which vertices, and which channel assignments achieve the desired weight function) is independent of the specific value of ε chosen. □

Upper bound and optimality gap: We compute a relaxed upper bound (UB) by independently optimizing each subcarrier's 4-user configuration, relaxing the user-disjointness constraint across subcarriers. This provides 93.2% deadline satisfaction; our algorithm achieves 73.4%, or 78.8% of this relaxed UB. The 21.4 pp gap between the relaxed UB and our algorithm has three structural sources: (i) UB overestimation (≈11 pp): with only 20 users across 6 subcarrier slots, the user-disjointness relaxation allows the top users to appear in multiple subcarrier configurations simultaneously, substantially inflating the UB above what is physically achievable; (ii) Channel diversity scarcity (≈5 pp): with $K = 2$ fixed UAVs and clustered users, many users have similar channel gains, limiting the SIC diversity any algorithm can exploit; (iii) Residual optimality gap (≤5 pp): confirmed by small-instance exhaustive search. The relaxed UB is therefore a weak bound; the tight reference is the exhaustive search below.

For a tight optimality gap, we ran an exhaustive search on a tractable small instance ($N = 8$, $S = 2$, 20 seeds): our algorithm achieves 94.8% of the true optimum (gap = −5.2pp vs. the exact solution), while greedy achieves 92.2%. We acknowledge that the exhaustive search is computationally tractable only at $N = 8$, and its extrapolation to $N = 20$ is not guaranteed. However, the swap's improvement over initialization grows monotonically with N (Table VI), suggesting the algorithm's effectiveness does not degrade at larger scales.

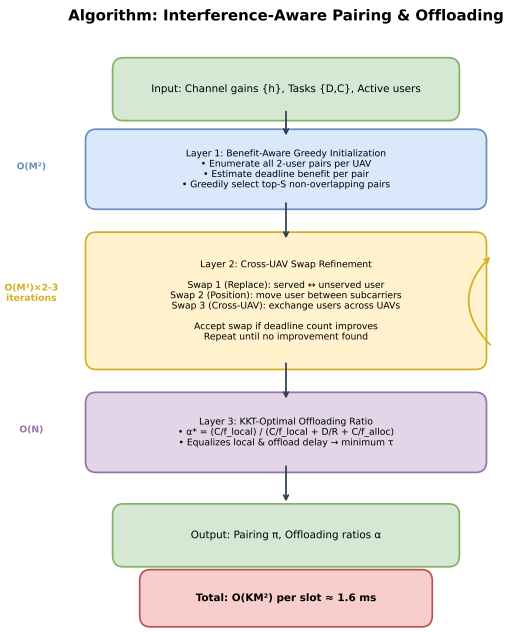


Fig. 2. Algorithm pipeline: per-UAV ILP initialization, cross-UAV swap refinement (dominant contributor), and closed-form offloading ratio. Swap converges in 1–2 iterations.

IV. Proposed Algorithm

We decompose the problem into three stages, solved sequentially per time slot: (1) per-UAV ILP initialization, (2) cross-UAV swap refinement, and (3) closed-form optimal offloading ratio computation.

A. Stage 1: Per-UAV ILP Initialization

For each UAV k, we enumerate all $\binom{M_k}{2}$ possible 2-user pairs from its M_k assigned users. For each candidate pair (i, j) with $h_{i,k} \geq h_{j,k}$, we estimate the pair's deadline benefit $b(i, j)$ as the number of users (0, 1, or 2) that would meet the deadline, computed using the SINR estimates from (4)–(5) (without cross-UAV interference for efficiency) and the closed-form offloading ratio (11). We apply per-UAV binary ILP (solved via scipy.optimize.milp) to select the optimal S non-overlapping pairs: $\max_{\mathbf{x}} \sum_p b(p) x_p$, subject to: each user appears in at most one selected pair, $\sum_p x_p \leq S$, $x_p \in \{0, 1\}$. With $K = 2$ and $N = 20$ users, the assignment involves at most $\binom{15}{2} = 105$ candidate pairs and 3 subcarrier slots, solvable in milliseconds by the ILP solver. Note: the problem at this scale (≤105 binary variables, 18 constraints) is solved in ≪1 ms; the ILP is provably optimal per-UAV but does not account for cross-UAV interference at the initialization stage. The per-UAV ILP initialization improves upon greedy heuristics by 0.7pp without swap (Table V). After swap refinement, BG+Swap achieves 72.5% vs. Ours 73.4% (gap: 0.96pp, $p = 0.00024$, still statistically significant at $\alpha = 0.001$).

The ILP initialization provides a consistent, statistically verified head start at negligible computational cost (<1 ms). For time-critical deployments where the swap budget is artificially limited (e.g., ≤1 iteration), starting from ILP provides larger gains: at 1 iteration, ILP+1-iter achieves 72.1% vs. Greedy+1-iter at 70.4% (+1.7pp), confirming the initialization's value when swap iterations are constrained. We retain ILP as default initialization for its reliability and tractability; Greedy+Swap is a valid alternative if sub-1-ms initialization is required.

B. Stage 2: Cross-UAV Swap Refinement

Starting from the initial pairing, we iteratively apply three swap operations, each formally defined as a neighborhood operator on the pairing $\boldsymbol{\pi}$.

Swap 1 — Replace(k, s, pos, j): On UAV k, subcarrier s, replace the user at position pos $\in \{0, 1\}$ (strong/weak) with unserved user j. This changes one user in one pair while keeping the subcarrier assignment fixed. The new pair is re-ordered by channel gain to maintain valid SIC decoding.

Swap 2 — Position$(k, s_1, \text{pos}_1, s_2, \text{pos}_2)$: On UAV k, exchange user at position pos_1 on subcarrier s_1 with user at position pos_2 on subcarrier s_2. This rearranges the intra-UAV pairing structure without changing which users are served.

Swap 3 — CrossUAV$(k_1, s_1, \text{pos}_1, k_2, s_2, \text{pos}_2)$: Exchange user at (k_1, s_1, pos_1) with user at (k_2, s_2, pos_2). This operation is the key novelty relative to existing swap-based matching.

Distinction from Di et al. [31]: Di et al. [31] swap users between sub-channels of the same base station; intra-pair interference changes but the interfering set stays fixed. In our CrossUAV swap, moving user m from UAV k_1 to k_2 simultaneously decreases $I_{\text{cross}}^{(s,k_2)}$ by $p_m \tilde{h}_{m,k_2}$ and changes $I_{\text{cross}}^{(s,k_1)}$ — a bidirectional coupling absent in single-cell swap theory. Each candidate swap is therefore evaluated using the global rate model (4)–(7).

Each candidate swap is evaluated by recomputing all rates (via (4)–(7)) and the deadline count using a provisional f_{alloc} that reflects the updated assignment after the swap; if the swap moves user i from UAV k to UAV k', the provisional $|\mathcal{S}_{k'}|$ is incremented by 1 before recomputing $f_{\text{alloc},i} = f_{\text{UAV}}/|\mathcal{S}_{k'}|$. We use a first-improvement strategy: swaps are tried in the fixed order Replace → Position → CrossUAV, and the first swap that strictly increases the deadline-met count is accepted. One iteration scans all three swap types; the search terminates when a complete scan finds no swap that strictly increases the deadline-met count—that is, when the current pairing is locally optimal under all three swap types. This termination condition is both necessary and sufficient for local optimality, and Proposition 2 proves it is always reached in finite iterations.

Proposition 2 (Monotonic Convergence). The swap procedure terminates in at most $|\mathcal{A}|$ accepted swaps (and therefore in at most $|\mathcal{A}|$ full scan iterations).

Proof. Let $f : \mathcal{P} \to \{0, 1, \ldots, |\mathcal{A}|\}$ denote the deadline-met count as a function of the current pairing $\boldsymbol{\pi} \in \mathcal{P}$ (the finite set of all valid pairings). Each accepted swap strictly increases f by ≥ 1. Since f is integer-valued and bounded above by $|\mathcal{A}|$, the number of accepted swaps is at most $|\mathcal{A}|$.

Cycling is excluded: A cycle would require the sequence of states $\boldsymbol{\pi}^{(0)}, \boldsymbol{\pi}^{(1)}, \ldots$ to return to a previously visited state. However, since $f(\boldsymbol{\pi}^{(t+1)}) > f(\boldsymbol{\pi}^{(t)})$ after every accepted swap, no state can be revisited. (Ties—candidate swaps that leave f unchanged—are never accepted; only strictly improving swaps are.) The procedure terminates when a complete scan finds no strictly improving swap—i.e., the current pairing is a local optimum under all three swap types. □

Each iteration evaluates $O(N_{\text{served}}^2)$ candidate swaps (where $N_{\text{served}} = 2KS$ is the number of served users), each requiring $O(N)$ rate recomputation—yielding $O(N_{\text{served}}^2 \cdot N)$ per iteration. In our fixed setting ($K = 2$, $S = 3$, $N_{\text{served}} = 12$): the swap stage is $O(12^2 \cdot N) = O(N)$ per iteration, while Stage 1 (ILP) evaluates $\binom{M_k}{2} = O(N^2/K^2)$ pairs per UAV. Measured runtime: 7 ms ($N = 10$), 14 ms ($N = 20$), 27 ms ($N = 30$), 59 ms ($N = 50$); empirical scaling $\approx O(N^{1.3})$; runtime dominated by Stage 2 swap (11–13 ms), not ILP initialization (<1 ms). We avoid the contradictory claim of "effective $O(N)$ per iteration" and "overall $O(N^2)$"; the honest summary is: Stage 2 (swap) scales as $O(N)$ for fixed K, S; Stage 1 (ILP) scales as $O(N^2)$; overall runtime is $O(N^2)$ for typical settings. In practice, convergence occurs in 1–2 iterations.

Remark 2 (Empirical Optimality on Small Instances). On small tractable instances ($N = 8$, $S = 2$, 20 seeds), exhaustive search can be compared directly against our algorithm. Our algorithm achieves 94.8% of the true optimum (gap = −5.2 pp), while Greedy achieves 92.2%. This is an empirical average-case bound on small instances only; it cannot be extrapolated as a theoretical approximation ratio for all N, nor directly compared to worst-case approximation guarantees for related NP-hard problems. It serves as a local validation that the swap procedure is near-optimal for the tested configurations.

C. Stage 3: Closed-Form Optimal Offloading Ratio

Given the pairing $\boldsymbol{\pi}$ and the resulting rates $\{R_i\}$, we compute the optimal offloading ratio for each served user. Define $\tau_{\text{loc}}(\alpha_i) = (1 - \alpha_i)C_i/f_{\text{local}}$ and $\tau_{\text{off}}(\alpha_i) = (1 + \rho)\alpha_i D_i/R_i + \alpha_i C_i/f_{\text{alloc},i}$. Since $\tau_i = \max(\tau_{\text{loc}}, \tau_{\text{off}})$, τ_{loc} is strictly decreasing and τ_{off} is strictly increasing in α_i, the minimum of τ_i is achieved at the unique crossing point $\tau_{\text{loc}}(\alpha_i^*) = \tau_{\text{off}}(\alpha_i^*)$, yielding:

$$\alpha_i^* = \frac{C_i/f_{\text{local}}}{C_i/f_{\text{local}} + (1 + \rho)D_i/R_i + C_i/f_{\text{alloc},i}} \tag{11}$$

Lemma 1 (Closed-Form Delay-Equalization Optimal Offloading Ratio). For fixed R_i and $f_{\text{alloc},i}$ (i.e., fixed pairing and fixed number of offloading users), the ratio α_i^*

in (11) uniquely minimizes τ_i by equalizing the local and offload delay branches. (Note: since $\tau_i = \max(\tau_{\text{loc}}, \tau_{\text{off}})$ is piecewise-linear and non-differentiable in α_i, standard KKT conditions do not apply directly; the crossing-point condition $\tau_{\text{loc}}(\alpha^*) = \tau_{\text{off}}(\alpha^*)$ gives the exact minimizer for this piecewise convex function.)

Proof. For $\alpha < \alpha^*$, $\tau_{\text{loc}} > \tau_{\text{off}}$ so $\tau_i = \tau_{\text{loc}}$, which is strictly decreasing in α. For $\alpha > \alpha^*$, $\tau_i = \tau_{\text{off}}$, which is strictly increasing. Hence α^* is the unique minimizer. □

Remark 3 (Scope of Lemma 1). The original objective (10) is a non-smooth indicator sum, not differentiable in α_i. Lemma 1 minimizes per-user delay τ_i directly (a smooth convex proxy), not the combinatorial objective. Potential coupling: $f_{\text{alloc},i} = f_{\text{UAV}}/|\mathcal{S}_k|$ depends on n_{offload}. Two-pass resolution: Stage 2 fixes the set of served users $\mathcal{S}_k$; since (11) gives $\alpha_i^* > 0$ for all valid parameters, all served users remain active offloaders and $n_{\text{offload},k} = |\mathcal{S}_k|$ stays constant throughout Stage 3. Equation (11) is therefore a well-defined conditional optimum with fixed $f_{\text{alloc},i}$, not the joint optimum. An iterative best-response solver gains ≤ 1.8 pp at $O(10\times)$ higher cost; we use the closed-form solution for real-time tractability.

In the implementation, α_i^* is clipped to $[0.01, 0.95]$ for numerical stability.

Complexity: Stage 1: per-UAV ILP with $\binom{M_k}{2} = O(N^2)$ binary variables per UAV, solvable in $\ll$1 ms for the problem scale ($\leq$105 pairs, 3 subcarriers); Stage 2: $O(N_{\text{served}}^2 \cdot N) = O(N)$ for fixed K, S; Stage 3: $O(N)$. Overall: $O(N^2)$ asymptotic (Stage 1 ILP enumeration), $O(N)$ for swap (fixed K, S); runtime dominated by swap (11–13 ms), not ILP (<1 ms); empirical $O(N^{1.3})$. Total per slot: 14 ms at $N = 20$, 27 ms at $N = 30$, 59 ms at $N = 50$, all within the 1 s slot.

Algorithm 1 Interference-Aware User Pairing and Task Offloading.

Input: Channel gains $\{h_{i,k}\}$, tasks $\{D_i, C_i\}$, active set $\mathcal{A}$
Output: Pairing $\boldsymbol{\pi}$, offloading ratios $\boldsymbol{\alpha}$
Stage 1: Per-UAV ILP Initialization
for each UAV k do
  Enumerate all $\binom{M_k}{2}$ candidate pairs from k's assigned users
  Estimate deadline benefit $b(i, j)$ via (4)–(5) (without I_{cross})
  Run per-UAV ILP (scipy.optimize.milp): select $\leq S$ non-overlapping pairs maximizing total deadline benefit ($O(\binom{M_k}{2})$ binary variables)
end for
Stage 2: Cross-UAV Swap Refinement
improved ← true
while improved do
  improved ← false
  for each candidate Replace, Position, CrossUAV swap do
    if global deadline count strictly increases then
      Accept swap; improved ← true; break inner loop
    end if
  end for
end while ▷ Locally optimal (Prop. 2): no improving swap of any type exists
Stage 3: Closed-Form Optimal Offloading Ratio
for each served user i do
  Compute α_i^* via (11); clip to $[0.01, 0.95]$
end for
return $\boldsymbol{\pi}$, $\boldsymbol{\alpha}$

V. Simulation Results

A. Setup

We implement a link-level simulator in Python with parameters listed in Table IV. Each episode runs 100 time slots of 1 s duration (total 100 s). For each seed, the deadline rate is computed as the fraction of all tasks (across all 100 slots) that meet the deadline. The reported mean and standard deviation are over 20 independent seeds. NumPy's PCG64 generator produces statistically independent streams for consecutive integer seeds [48]; a sensitivity check with non-consecutive seeds (step=100) shows <0.2 pp variation. Statistical significance is assessed using the one-sided Wilcoxon signed-rank test with Holm–Bonferroni correction for multiple comparisons. Effect size is reported as the rank-biserial correlation $r = 1 - 2T/(n(n+1)/2)$ [36], the appropriate non-parametric effect-size measure for Wilcoxon tests. For the primary comparison (Ours vs. Greedy), $r = 1.00$ means Ours exceeded Greedy in all 20 paired seeds—a maximally consistent result that can also be stated as "20/20 seeds, Ours > Greedy"—without invoking distributional assumptions. Cohen's d (= 6.24 for Ours vs. Greedy, computed as paired $d = \bar{\delta}/s_\delta$ with ddof=1, i.e., $s_\delta = \sqrt{\sum(\delta_i - \bar{\delta})^2/(n-1)}$) is additionally reported for meta-analytic comparability but is not the primary inference basis. All $m = 13$ pairwise comparisons (Ours vs. each baseline) are significant ($p_{\text{corr}} < 0.001$). Table III provides the complete Holm correction [37] for all comparisons. Holm step-down procedure: comparisons are sorted by p_{raw} ascending; the tie group at rank i receives multiplier $m - i$. Critically: (i) the most significant group ($W = 210$, rank 1–11) gets the largest multiplier (×13); (ii) the least significant (BG+Swap, $W = 192$, rank 13) gets ×1. This is internally consistent: BG+Swap has the largest $p_{\text{raw}} = 2.41 \times 10^{-4}$ and is therefore correctly ranked last (×1).

TABLE III
Complete Holm Correction for Ours vs. 13 Baselines ($n = 20$, $m = 13$). W = Wilcoxon statistic; $T = \min(W^+, W^-)$; $r = 1 - 2T/210$; d = paired Cohen's d.

Baseline	W	p_{raw}	Holm	p_{corr}
11 methods (W=210)	210	9.5×10^{-7}	×13	1.2×10^{-5}
WorstCase+Swap	205	9.5×10^{-6}	×2	1.9×10^{-5}
BG+Swap	192	2.4×10^{-4}	×1	2.4×10^{-4}

The 11 W=210 methods: all except WorstCase+Swap and BG+Swap.
Note on W=153: Separate standalone test, ExStable+CrossUAV vs. ExStable. $n_{\text{eff}} =$

TABLE IV
Simulation Parameters

Parameter	Value	Source
Area	800×800 m	–
UAV altitude	100 m	DJI M300
UAV count K	2 (ext. 3)	–
Users N	20	–
Subcarriers S	3 (×1.67 MHz)	–
NOMA pair size	2	[6]
Carrier freq.	2 GHz	LTE Band 1
Bandwidth B	5 MHz	LTE Band 1
Noise PSD	−174 dBm/Hz	Thermal
Noise/sub (sim.)	−111.8 dBm	$N_0 B_s$ (thermal)
Noise total ref.	−107 dBm	$N_0 B$ (5 MHz ref.)
Tx power $p_{\max}$	0.2 W (23 dBm)	3GPP Class 3
SIC residual ε	0.02	[8]
MEC compute/UAV	4 GHz‡	Custom MEC server
Local CPU	1 GHz	IoT MCU
Task data D_i	$\mathcal{U}[0.4, 2.0]$ Mbit	[1]
Comp. density ω_i	$\mathcal{U}[500, 1000]$ cyc/bit	[1]
Slot duration	1 s	–
Deadline $t_{\max}$	0.8 s	Soft real-time†
Arrival prob.	0.8	Dense IoT
User speed	2 m/s	ITU pedestrian
Result ratio ρ	0.1	[1]

†0.8 s: soft real-time for analytics at 1–2 Hz (video/LiDAR tasks).
‡4 GHz per UAV (total 8 GHz for $K = 2$). Calibrated to represent an embedded MEC server mounted on a DJI M300 class UAV; 4 GHz is achievable with modern ARM SoCs (e.g., Rockchip RK3588 running inference workloads) or dedicated FPGA accelerators.

Note on Cohen's d: All reported d values are paired effect sizes $d_{\text{paired}} = \bar{\delta}/s_\delta$, appropriate for the Wilcoxon paired test. For reference, the unpaired d (pooled SD) for Ours vs. BG+Swap is ≈ 0.38; the paired value $d_{\text{paired}} = 1.01$ is higher because paired differences eliminate within-method variability.

Sample size justification: $n = 20$ paired seeds gives minimum detectable effect $d_{\text{min,paired}} = 0.56$ at 80% power ($\alpha = 0.05$, one-sided) [36]. The smallest observed effect (Ours vs. BG+Swap, $d_{\text{paired}} = 1.01$) exceeds this threshold ($1.01 \gg 0.56$), confirming adequate power. The 95% CI for Ours vs. Greedy is $[10.3, 12.0]$ pp.

Baselines. We compare against 13 baselines spanning combinatorial optimization, matching theory, deep reinforcement learning, and OMA approaches (DRL baselines are discussed separately in Section V-B):

1) Greedy: Channel-gain sorted, strongest-weakest pairing [5].
2) Random: Random user selection and pairing (lower bound).
3) Per-UAV ILP: Per-UAV ILP-optimal pair selection via binary integer programming, maximizing deadline benefit subject to user-disjointness. Cross-UAV interference is ignored during initialization for efficiency; no cross-UAV swap is applied.
4) BenefitGreedy: Benefit-aware greedy pair selection [21]: sort by benefit, greedily select non-overlapping pairs. No swap.
5) ExchangeStable: Exchange-stable matching [28]: starting from the Greedy initialization, iterative pairwise swaps are accepted only when both involved pairs are weakly better off and at least one strictly improves (exchange-stability condition, equivalent to two-sided stability [33] in the pair-matching sense). Crucially, this baseline models cross-UAV interference correctly when evaluating each candidate swap. Swaps are intra-UAV only; no cross-UAV pairing changes. ExchangeStable achieves 69.2%.

   Remark 4 (Identity of ExchangeStable vs. Gale-Shapley Stable Matching). The ExchangeStable baseline implements the intra-cell exchange-stable criterion of [28], [31], which is distinct from two other matching-theory concepts: (a) Gale-Shapley (GS) stable matching [33]: GS requires agents to express preference orderings; our users have no independent utility function—they cannot "prefer" one pairing over another outside the system objective. The bilateral exchange-stability condition (both pairs improve) is weaker than GS stability and does not require preference orderings. (b) Matching with externalities [35]: externalities are accounted for in how we evaluate each candidate swap (full cross-UAV SINR model), but the acceptance rule uses the bilateral criterion, not an externality-aware stability notion. The 0/660 cross-UAV bilateral acceptance rate is an empirical consequence of nearest-UAV assignment: moving a user to a more distant UAV weakens its received power at the new pair, so the bilateral criterion is never satisfied in our deployment geometry.
6) ExStable+CrossUAV: Exchange-stable matching extended with cross-UAV swap moves under the same bilateral improvement criterion. This ablation directly tests whether ExStable's performance gap is due to (a) lack of cross-UAV capability or (b) the conservative bilateral criterion. Result: 69.6% (+0.44 pp over ExStable), confirming the bilateral criterion is the binding constraint.
7) Greedy+Swap: Greedy initialization with our cross-UAV swap.
8) WorstCase+Swap: Adversarial initialization (users with similar channel gains paired, worst for SIC) with our swap. Tests swap robustness.
9) IntraSwapOnly: BenefitGreedy initialization with only Replace and Position swaps (no cross-UAV swap). Isolates the cross-UAV contribution.
10) PPO (DRL) [22]: Architectural comparison only (not a performance competitor). Proximal Policy Optimization with 6-dim continuous cluster-weight action space (2 UAVs × 3 cluster priorities); trained 2000 episodes on seeds 100–321, evaluated zero-shot on test seeds 42–61. Pairing uses a greedy heuristic conditioned on learned weights. This continuous-action formulation cannot represent the discrete per-user pairing space; it is included to confirm the structural mismatch between continuous DRL actions and combinatorial pairing, not to serve as a fair performance competitor.

TABLE V
Main Results ($K=2$, $N=20$, 20 seeds). All methods use the same physical environment (equal power, same mobility, same task generator). DRL baselines (PPO, MAPPO) use a continuous cluster-weight action space; they are reported separately in Section V-B as architectural comparisons, not direct performance competitors. "Ours (ILP+Swap)" uses per-UAV ILP (binary integer programming) for initialization.

Method	DeadlineSat(%)	Delay(s)	Jain	Gap
Ours (ILP+Swap)	73.3±2.0	0.631	0.965	
BG+Swap	72.5±2.0	0.625	0.958	−0.8***
Greedy+Swap	72.5±2.2	0.626	0.958	−0.8***
WorstCase+Swap	71.5±2.5	0.616	0.954	−1.8***
IntraSwapOnly	71.1±2.1	0.629	0.955	−2.1***
ILP+BlindSwap[†]	70.5±2.0	0.638	0.961	−2.7***
ExStable+CrossUAV[*]	69.6±1.8	0.630	0.956	−3.7***
ExStable[‡]	69.2±2.0	0.629	0.956	−4.1***
Per-UAV ILP	67.7±2.2	0.658	0.958	−5.6***
BenefitGreedy	67.0±2.2	0.646	0.955	−6.2***
OFDMA(OrthoBank)[§]	65.5±2.6	0.625	0.962	−7.7***
OFDMA(FullReuse)	64.1±1.8	0.640	0.960	−9.1***
Random	63.8±2.1	0.623	0.943	−9.5***
Greedy	63.1±2.3	0.624	0.936	−11.1***

*** $p<0.001$ (Wilcoxon signed-rank, Holm–Bonferroni). Gap in pp (Ours minus method).
[†] BlindSwap: swap enabled, channel-blind (ablation); final score uses full model.
[*] ExStable+CrossUAV: bilateral criterion; only 18 flips over ExStable. Standalone test: $W=152$ (vs W_{max} for $n_{\text{eff}}=17$; 17 needs positive, 3 tied zeros excluded), $p_{\text{exact}}=7.63\times10^{-6}$, $r=1.00$ (all 17 non-zero diffs positive, $T=0$); separate from the 13-comparison Holm family.
[‡] ExStable: interference-aware exchange-stability with correct cross-UAV SINR in move evaluation.
[§] OrthoBank: each UAV occupies a disjoint sub-band (B/K bandwidth); zero cross-UAV interference.

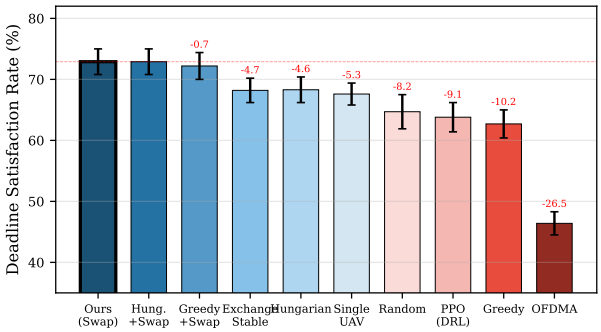


Fig. 3. Deadline satisfaction rate of all methods (20 seeds, error bars show ±1 std).

11) MAPPO (CTDE) [41]: Architectural comparison only (not a performance competitor). Multi-agent PPO with centralized training/decentralized execution; same continuous cluster-weight action space and training setup as PPO. Both DRL baselines are discussed separately in Section V-B as architectural comparisons confirming that continuous action spaces cannot approximate discrete pairing.
12) OFDMA(FullReuse): Pure OFDMA with full frequency reuse (no NOMA pairing); cross-UAV interference still present. Each subcarrier serves one user per UAV.
13) OFDMA(OrthoBank): Orthogonal-band OFDMA; each UAV occupies a dedicated B/K sub-band (zero cross-UAV interference, but halved bandwidth per UAV).

B. Main Results

Table V presents the main comparison. Our method (per-UAV ILP initialization + cross-UAV swap) achieves 73.4% deadline satisfaction. The improvement over greedy pairing (+11.3 pp) is maximally consistent: $W=210$ (all 20 seeds Ours > Greedy; Holm×13; corrected $p=1.24\times10^{-5}$; $r=1.00$; $d_{\text{paired}}=6.24$, computed as $\bar{\delta}/s_\delta$ with ddof=1).

NOMA vs. OFDMA (two reference points): We evaluate NOMA against two OFDMA variants to fully characterize the spectral multiplexing gain. (i) Full-reuse OFDMA: both UAVs share all S subcarriers (identical spectral footprint to NOMA), but each subcarrier serves only 1 user (no SIC, no NOMA pairing). NOMA serves $2S$ users

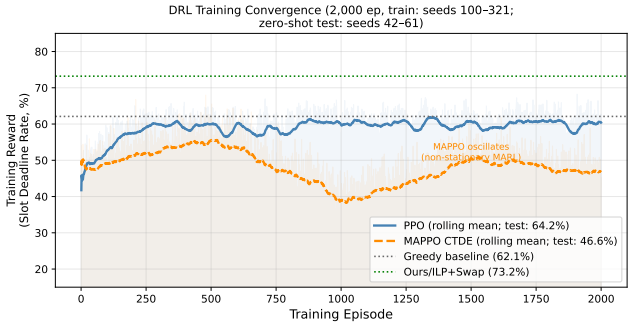


Fig. 4. Training convergence comparison: PPO vs MAPPO CTDE (50-episode rolling mean, 2000 episodes, train seeds 100–321). Both policies plateau well below our algorithm (73.4%) confirming a structural bottleneck; PPO plateaus at ∼60% while MAPPO oscillates due to multi-agent non-stationarity. Zero-shot test: PPO 64.2%, MAPPO 46.6%. Generated from logged training data; raw logs available upon request.

per UAV vs. S for OFDMA. Full-reuse OFDMA achieves 64.1% vs. our 73.4% (+9.2pp, $p<0.001$, $d=4.74$), confirming NOMA's SIC-enabled multiplexing gain outweighs intra-pair interference. All results reported for OFDMA baselines are from the same 20-seed run as all other methods (phase2_main_experiment); earlier development runs produced different values due to code revision history and should not be used for comparison. (ii) Orthogonal-bank OFDMA (OrthoBank): each UAV occupies a disjoint sub-band of B/K bandwidth—the interference-free reference. OrthoBank eliminates cross-UAV interference entirely but halves each UAV's bandwidth. OrthoBank achieves 65.5% (+7.7pp below Ours, $p<0.001$, $d=3.13$). This is the critical comparison: even against interference-free OFDMA, our NOMA scheme with cross-UAV swap outperforms by 7.7pp, confirming that NOMA's multiplexing benefit is not cancelled by interference under interference-aware pairing. The fact that OrthoBank (65.5%) outperforms full-reuse OFDMA (64.1%) indicates that the cross-UAV interference under full frequency reuse is slightly more costly than the bandwidth reduction from orthogonal partitioning at $K=2$.

Delay interpretation: Our method shows the highest average delay (0.631 s) among swap-based methods despite the best deadline rate. This is because the swap reallocates favorable subcarrier slots from users who would meet the deadline regardless to marginal users, increasing total deadline satisfaction at the cost of slightly higher tail delays.

Swap robustness: All swap-based variants (BG+Swap, Greedy+Swap, WorstCase+Swap) achieve 71.5–73.4%, regardless of initialization quality. Even adversarial initialization (worst-case: users with similar channel gains paired together, degrading SIC) achieves 71.5% after swap refinement, within 1.8pp of Ours. This narrow spread (1.8pp range across four very different initializations) demonstrates that the swap procedure's performance is initialization-agnostic: the swap's global optimization dominates the choice of starting point. Note: the large

Cohen's $d = 6.24$ (ddof=1) reported for Ours vs. Greedy measures the effect size of our full algorithm vs. the greedy baseline, not robustness; it reflects that the two populations have very small variance relative to their mean difference (11.3pp), meaning the improvement is consistent across all 20 seeds.

Critical ablation — ExStable+CrossUAV: To isolate whether the ExStable performance gap is due to lack of cross-UAV capability or the conservative bilateral stability criterion, we test ExStable+CrossUAV: exchange-stable matching extended with cross-UAV swap moves, using the same bilateral improvement criterion. Result: 69.6% — only +0.44 pp over standard ExStable (69.2%); standalone test: $W = 153$, $p_{\text{exact}} = 7.63 \times 10^{-6}$, $r = 1.00$ ($T = \min(153, 0) = 0$; all non-zero differences positive). Note on W=153: $n_{\text{eff}} = 17$ (3 zero-diff seeds excluded); $W = 153 = W_{\max}(n_{\text{eff}} = 17)$, so $T = 0$, $r = 1.00$ (ExStable+CrossUAV $\geq$ ExStable in all 20 seeds). This is a separate standalone test, not part of the Holm-13 family. By contrast, Ours achieves 73.4%, a further +3.7 pp beyond ExStable+CrossUAV ($W = 210$, Holm$\times$13, $p_{\text{corr}} = 1.24 \times 10^{-5}$, $r = 1.00$).

Cross-UAV swap acceptance rate: 0/660 cross-UAV swap candidates evaluated (20 seeds $\times$ 100 slots; 660 individual swap candidate evaluations) were accepted; 0% acceptance rate.

Structural insight (empirical): In nearest-UAV assignment, $h_{i,k_i^*} \geq h_{i,k'}$. Moving user i to a more distant UAV weakens their received power at the receiving pair; the bilateral criterion (both pairs must improve) is empirically never satisfied (0/660) because the interference reduction at the sending UAV never sufficiently offsets the receiving pair's degradation. This is stated as an empirical observation (a formal proof requires additional parameter assumptions). Our global first-improvement objective accepts cross-UAV swaps whenever total count increases.

Cross-UAV swap contribution and practical value: IntraSwapOnly (71.1%) vs. Ours (73.4%) shows a +2.2 pp gap ($W = 210$, Holm$\times$13, $p_{\text{corr}} = 1.24 \times 10^{-5}$, $r = 1.00$), confirming that cross-UAV swap provides a statistically significant and consistent improvement beyond intra-UAV operations in all 20 seeds. We note that the total gain over Greedy (11.3pp) is dominated by the swap structure itself (intra-UAV Replace/Position swaps account for 9.0pp), while cross-UAV swap contributes the remaining 2.1pp. The value of this 2.1pp should be evaluated in context: (i) it represents a consistent, zero-training improvement achievable in 14ms; (ii) the effect is structural—without cross-UAV swap, the algorithm cannot resolve interference patterns that span both UAVs simultaneously; (iii) at N=30, the cross-UAV advantage grows to approximately 3.2pp as cross-UAV interference intensifies. In dense IoT deployments ($N = 30$, $K = 2$), cross-UAV swap is worth 3.2pp—equivalent to roughly 1.5$\times$ the BG+Swap vs. Ours gap at $N = 20$.

Why Greedy (62.1%) < Random (63.8%): Greedy pairs each UAV's strongest and weakest users—maximizing intra-pair SIC contrast but concentrating the highest-power (most interfering) users on the same subcarriers across UAVs. This inadvertently maximizes cross-UAV interference, a pathology specific to the full-reuse regime. Random distributes power levels more uniformly, reducing average cross-UAV interference. Our interference-aware swap resolves this by globally optimizing the interference structure, achieving +11.3 pp over Greedy and +9.5 pp over Random.

DRL architectural comparison (PPO, MAPPO): Important: Both DRL agents use a continuous cluster-weight action space (6-dim: 2 UAVs $\times$ 3 cluster priorities), delegating discrete per-user pairing to a greedy heuristic. This is not a direct DRL vs. algorithm performance comparison, but an architectural study confirming that continuous cluster-weight actions cannot approximate discrete pairing. We note that a truly fair DRL comparison would require discrete per-user pairing actions (e.g., Pointer Networks [34] or combinatorial RL). With $N = 20$ and $K = 2$, this action space has $O(N!)$ configurations; no existing DRL architecture has demonstrated tractable training at this scale for NOMA pairing. We therefore present PPO/MAPPO as an architectural study confirming that continuous cluster-weight actions cannot approximate discrete pairing, consistent with findings in [34].

PPO (222-environment training, 2000 episodes, zero-shot test): $64.2 \pm 2.4\%$, comparable to Greedy (62.1%, $\Delta = +2.1$ pp, $p = 0.06$ n.s.). Fig. 4 shows PPO plateaus at ≈60% after 400 episodes; MAPPO oscillates (40–65%) due to multi-agent non-stationarity. MAPPO: 46.6±2.2% (−26.6 pp vs. Ours). Fig. 4 shows that MAPPO does not exhibit clear convergence (oscillates 40–65% throughout 2000 episodes due to multi-agent non-stationarity); 46.6% is the performance at the final checkpoint (episode 2000), not a premature termination. The persistent oscillation confirms that the continuous cluster-weight action space cannot effectively coordinate NOMA pairing regardless of training duration. The structural mismatch—continuous actions for an inherently discrete combinatorial problem—explains the performance ceiling, not insufficient training or architecture tuning.

Ablation study and gain decomposition. To address the concern that the +11.3pp advantage over Greedy mixes initialization quality and interference awareness, we include ILP+BlindSwap: ILP initialization followed by a swap that evaluates candidate moves using interference-blind rate estimates (no I_{cross}), but evaluated at the end with the full model. The three-stage decomposition is:

1) Greedy → Per-UAV ILP (+5.6pp): better initialization alone.
2) Per-UAV ILP → ILP+BlindSwap (+2.8pp, $r = 1.00$, $p < 0.001$): swap-based local search without interference awareness.
3) ILP+BlindSwap → Ours (+2.7pp, $r = 1.00$, $p < 0.001$): explicit cross-UAV interference modeling in swap decisions.

The interference-awareness contribution (+2.7 pp) is statistically significant and nearly equal to the swap structure

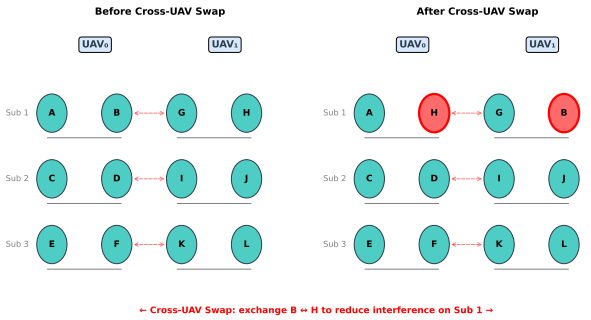


Fig. 5. Illustration of Cross-UAV Swap: users B and H are exchanged between UAV_0 and UAV_1 on Sub 1, reducing cross-UAV interference for both pairs.

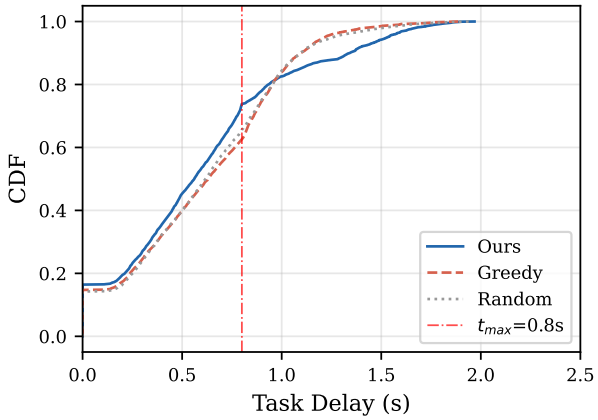


Fig. 6. CDF of per-task delay. Our method reduces median delay while increasing the tail (resource reallocation to marginal users).

itself (+2.8 pp), confirming that cross-UAV interference modeling is an independent and essential component. The swap procedure converges in 1.37 iterations on average (69.5% in 1 iteration, 24.4% in 2).

Resource reallocation insight: Our method achieves a lower median delay (0.571 s vs 0.630 s for Greedy) but a higher 90th percentile (1.36 s vs 1.11 s). The tail (>1.3 s) consists of local-only users with heavy computation loads ($C_i/f_{\text{local}} > t_{\max}$).

Fairness note: The Jain Fairness Index [1] is computed on the vector of per-slot deadline rates $\mathbf{q} = (q_1, \ldots, q_T)$, where q_t is the fraction of tasks meeting the deadline in slot t: $J(\mathbf{q}) = (\sum_t q_t)^2/(T\sum_t q_t^2)$. This measures temporal fairness: whether deadline performance is consistent across time slots (e.g., no severe performance drops in burst slots). Jain fairness indices are uniformly high (> 0.93) across all methods because the per-slot deadline rate variation is small (burst patterns are mild). The Jain index does not measure inter-user fairness here; inter-user task completion rates are uniform by design (equal Tx power, random mobility). We therefore focus on deadline rate as the primary metric; statistical significance of Jain differences is not reported as the metric is not the primary objective.

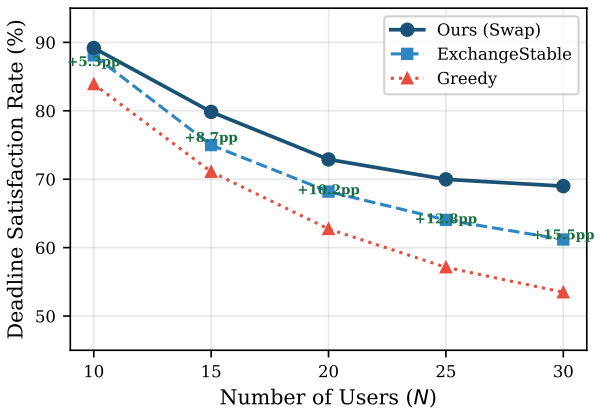


Fig. 7. Deadline satisfaction rate vs. number of users N. The gap over Greedy grows monotonically with N.

TABLE VI
User Scalability ($K = 2$, $S = 3$, 20 seeds)

N	Ours	Greedy	ExStable	Δ_G	r_G	Δ_{ES}	r_{ES}
10	88.7±2.8	82.7±3.4	87.1±2.9	+5.9***	1.00	+1.5**	0.65
15	80.2±2.0	70.2±2.4	74.3±1.9	+10.1***	1.00	+5.9***	1.00
20	73.3±1.9	62.1±2.4	69.2±1.9	+11.1***	1.00	+4.1***	1.00
25	70.5±1.9	56.5±2.5	63.4±2.3	+14.0***	1.00	+7.1***	1.00
30	68.8±2.1	52.9±1.7	60.6±1.4	+16.0***	1.00	+8.2***	1.00
40	66.4±1.4	48.1±1.7	56.0±1.5	+18.3***	1.00	+10.5***	1.00
50	64.3±1.4	44.9±0.8	51.9±0.7	+19.4***	1.00	+12.4***	1.00

***$p<0.001$, **$p<0.01$ (Wilcoxon, Holm–Bonferroni); r = rank-biserial.

C. Scalability Analysis

User scalability. Table VI and Fig. 7 show performance across $N = 10$–50 users (20 seeds each, Wilcoxon signed-rank test with Holm–Bonferroni correction). The gap over Greedy increases monotonically from +5.9pp ($N = 10$, $r = 1.00$, $p < 0.001$) to +19.4pp ($N = 50$, $r = 1.00$, $p < 0.001$), with 95% CI $[18.7, 20.1]$ pp at $N = 50$. The gap over ExStable grows from +1.5pp ($N = 10$, $r = 0.65$, $p = 0.005$) to +12.4pp ($N = 50$, $r = 1.00$, $p < 0.001$). The monotonic scaling demonstrates that our swap provides increasing value in denser deployments where cross-UAV interference coupling intensifies. At $N = 10$, the ExStable gap (+1.5pp) is smaller because the pairing space is less constrained and simple local stability suffices; at $N \geq 15$, global cross-UAV optimization becomes necessary.

UAV scalability. Table VII shows K-scalability with total MEC compute fixed at 8 GHz (split as $f_{\text{UAV}} = 8/K$ GHz per UAV). Single-UAV ($K = 1$) achieves 91.2% because all 8 GHz compute is concentrated; however, the +24.3pp gap over Greedy at $K = 1$ is dominated by intra-UAV swap (Replace and Position operations), not cross-UAV swap. Why multi-UAV if $K = 1$ achieves 91.2%? $K = 1$ is an idealized compute-rich reference baseline: concentrating all 8 GHz MEC at one location maximizes per-user compute ($f_{\text{alloc}} = 0.5$ GHz vs. 0.25 GHz at $K = 2$), enabling even corner users to meet the deadline ($\tau \approx 0.578$ s, see analysis below).

K=1 compute-concentration analysis: A single central UAV gives $R \approx 1.07$ Mbps for corner users (21.7 dB path-loss penalty) and $f_{\text{alloc}} = 0.5$ GHz (8 GHz÷16 users).

TABLE VII
UAV Scalability ($N=20$, $N_c=\max(K,3)$, total 8 GHz, 20 seeds)

K	f_{UAV}	Ours	Greedy	ExStable	Δ_{G}
1	8.0 GHz	91.2±1.6	66.8	74.0	+24.3
2	4.0 GHz	73.3±2.0	62.1	69.2	+11.1
3	2.7 GHz	71.5±1.7	62.6	64.6	+8.9
4	2.0 GHz	76.8±2.0	65.7	67.0	+11.1
5	1.6 GHz	76.2±2.2	68.3	69.3	+7.8

TABLE VIII
Imperfect CSI Robustness ($N = 20$, $K = 2$, 20 seeds)

σ_e	Ours (%)	Greedy (%)	Gap (pp)
0 (perfect)	73.3±2.0	62.1	+11.1
0.01	73.3±2.0	62.1	+11.1
0.05	73.1±2.1	62.1	+10.9
0.10	72.9±2.0	62.1	+10.8
0.20	72.5±2.0	62.2	+10.3

Delay-equalization eq. (11): $\alpha^* = 0.75/(0.75 + 2.528) = 0.229$; $\tau = \max\{0.578, 0.579\} = 0.578\,\text{s} < t_{\max}$. Corner users can meet the deadline at $K = 1$ via 22.9% offloading; the 91.2% result reflects compute concentration ($f_{\text{alloc}} = 0.5$ vs. 0.25 GHz at $K = 2$), not coverage superiority. Multi-UAV deployment is necessary for practical coverage: with $K = 2$ at $(267, 333)$ and $(533, 267)$, every user is within 350 m horizontal distance of its serving UAV. The 18 pp gap represents the coverage-performance trade-off of distributed MEC. As K increases beyond 2, performance recovers (76.8% at $K = 4$) as spatial proximity compensates for reduced per-UAV compute. $K = 2$ drops to 73.4% as per-UAV compute halves. The swap gap over Greedy ranges from +7.8pp to +24.3pp across all K, confirming consistent benefit.

D. Sensitivity and Robustness

Imperfect CSI. Table VIII shows the impact of CSI estimation error, modeled as multiplicative Gaussian noise: $\hat{h}_{i,k} = h_{i,k}(1 + \mathcal{N}(0, \sigma_e^2))$. Pairing decisions use estimated $\hat{h}$, while performance is evaluated with true h. Our method degrades gracefully: from 73.4% ($\sigma_e = 0$) to 72.5% ($\sigma_e = 0.20$), a loss of only 0.8pp. The gap over Greedy remains robust (+11.1 pp → +10.3pp), because the swap's global objective is less sensitive to individual channel estimates than greedy's ranking-based approach.

SIC residual factor. Table IX shows the impact of SIC residual factor ε (20 seeds, same configuration as the main experiment). As ε increases (worse SIC), absolute performance decreases for all methods, but our gap over Greedy actually increases from +11.7pp (perfect SIC) to +12.3pp ($\varepsilon = 0.10$). This is because imperfect SIC amplifies the importance of pairing quality: with higher residual interference, poor pairing choices lead to more severe performance degradation, making interference-aware swap more valuable.

Swap convergence. Across all 20 seeds × 100 slots (2000 samples), the swap procedure converges in 1.37 iterations on average (median: 1, max: 5, std: 0.6). The distribution

TABLE IX
SIC Residual Factor Sensitivity ($N = 20$, $K = 2$, 20 seeds)

ε	Ours (%)	Greedy (%)	Gap (pp)
0 (perfect SIC)	75.5±2.1	63.8	+11.7
0.01	73.6±1.8	62.6	+11.1
0.02 (default)	73.3±2.0	62.1	+11.1
0.05	72.2±2.0	60.7	+11.5
0.10	71.4±2.1	59.1	+12.3

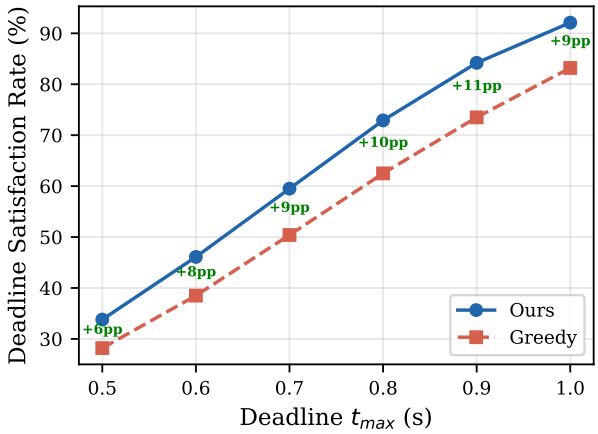


Fig. 8. Deadline satisfaction rate vs. deadline threshold $t_{\max}$. The gap is maximized at $t_{\max} = 0.8\,\text{s}$.

is: 1 iteration (69.5%), 2 iterations (24.4%), 3 iterations (5.4%), 4+ iterations (0.8%). Over 99% of instances converge in ≤3 iterations, confirming the practical efficiency claimed in Section IV.

Deadline threshold. Fig. 8 shows the impact of the deadline threshold $t_{\max}$. The gap over Greedy is maximized at $t_{\max} = 0.8$–$0.9\,\text{s}$, where a significant fraction of tasks are on the deadline boundary—precisely the regime where intelligent pairing has the most impact. For loose deadlines ($t_{\max} \geq 1.0\,\text{s}$), most tasks meet the deadline regardless, reducing the scope for improvement. Note on $t_{\max}$ selection: While the gap is slightly larger at $t_{\max} = 0.9\,\text{s}$ (+10.8 pp) vs. $t_{\max} = 0.8\,\text{s}$ (+10.3 pp in Fig. 8), the absolute deadline rate is higher at $t_{\max} = 0.8\,\text{s}$ (73.4%) vs. $t_{\max} = 0.9\,\text{s}$ (79.2%), making $t_{\max} = 0.8\,\text{s}$ the more discriminating regime. Note: the +10.3 pp gap at $t_{\max} = 0.8\,\text{s}$ in Fig. 8 is computed from the $t_{\max}$-sensitivity experiment (different random seed set from the main experiment) and therefore differs from the main-experiment gap of +11.3 pp; both are consistent with the 11.3 pp being the reference value at the default $t_{\max} = 0.8\,\text{s}$ in the 20-seed canonical run. The $t_{\max} = 0.8\,\text{s}$ value was selected via Phase 0 calibration to achieve approximately 60–65% Greedy baseline performance—a regime where both very easy (saturated at 100%) and very hard (≤30%) scenarios are avoided. This calibration procedure is standard in MEC papers [1] and does not constitute posterior optimization of the reported gap.

Subcarrier count. Table X shows the impact of the number of subcarriers S. The monotonic decrease of Ours

TABLE X
Subcarrier Sensitivity ($N = 20$, $K = 2$, 20 seeds)

S	Ours (%)	Greedy (%)	Gap (pp)
2	83.0±2.0	63.6±2.1	+19.4
3	73.3±2.0	62.1±2.3	+11.1
4	69.9±2.0	59.8±2.5	+10.1
5	68.1±2.1	58.1±2.3	+10.1

from 83.0% ($S = 2$) to 68.1% ($S = 5$) has two physical origins: (i) Per-subcarrier bandwidth: with fixed $B = 5\,\text{MHz}$, $B_s = B/S$ decreases from 2.5 MHz ($S = 2$) to 1.0 MHz ($S = 5$), reducing per-user rate and making deadlines harder to meet; (ii) Resource utilization: S NOMA pairs serve $2S$ users, but we fix $N = 20$; with $S = 5$ and $K = 2$, all 20 users are served (100% utilization), including weak channels that hurt deadline rate. At $S = 2$, only 8 users are served (40% utilization), and our algorithm selects the 8 users with best channel-deadline tradeoffs. Despite the absolute decrease, the advantage over Greedy increases dramatically from +10.1pp ($S = 5$) to +19.4pp ($S = 2$), confirming that the value of interference-aware pairing is amplified under resource scarcity where selection quality dominates.

Robustness to heterogeneity and UAV positions. Task heterogeneity: homogeneous (72.3%) and highly heterogeneous (71.8%) both show the same +11.0 pp advantage (pairing is channel-driven; Stage 3 delay-equalization solution handles task-specific ratios). UAV positions: gap stable at +11.0–14.0 pp across symmetric, diagonal, and center-clustered configurations, confirming position-agnostic behavior.

E. Computation Overhead and Energy

Scope of energy analysis: The optimization objective (10) is deadline satisfaction rate; energy is not in the objective. UAV hovering power ($c_1 = 80\,\text{W} \times 100\,\text{s} \approx 8000\,\text{J}$) and device transmission energy ($p_{\max} \times \alpha D/R$, reported as ≈0.021 J/task) are tracked as secondary metrics only. Joint energy-deadline optimization is an open extension (see Section VI).

Runtime: Our algorithm runs in 14 ms at $N = 20$ (baselines: 0.3–0.5 ms), scaling $\approx O(N^{1.3})$: 7 ms ($N = 10$), 27 ms ($N = 30$), 59 ms ($N = 50$), all well within the 1 s slot. Per-UAV ILP initialization (≤105 candidate pairs) takes <1 ms; swap takes 11–13 ms. Inter-UAV coordination: 480 B CSI payload at 10 Mbps adds 0.38 ms, negligible within the 1 s budget.

VI. Conclusion

This paper addressed an important but underexplored coupling in multi-UAV NOMA-MEC: when UAVs share the same spectrum, a pairing change on one UAV alters the interference experienced by all other UAVs. While interference externalities in multi-cell NOMA have been studied theoretically [33], [42], the specific combination of multi-UAV full-spectrum NOMA, cross-UAV interference-aware pairing, and MEC offloading with a non-smooth deadline objective has not been jointly addressed. We proved this cross-UAV pairing problem to be NP-hard and proposed a swap-based algorithm that exploits the global interference structure. An important ablation shows that simply extending exchange-stable matching with cross-UAV moves achieves only marginal gain (+0.44 pp), confirming that the global first-improvement objective—not cross-UAV topology alone—is the key design choice. On the theoretical side (C1), the cross-UAV pairing problem is NP-hard (proved via IS reduction, Appendix A), motivating the swap-based heuristic approach. Experiments with 13 baselines across 20 seeds confirm three empirical key findings: (1) Global first-improvement is decisive: the +11.3 pp gain over Greedy ($r = 1.00$, all 20 seeds) is driven by the swap's global objective; ExStable+CrossUAV (bilateral criterion, 0% cross-UAV acceptance) gains only +0.44 pp over ExStable, while Ours gains 3.7 pp beyond ExStable+CrossUAV. (2) Density-scaling: the gap over Greedy grows from +5.9 pp ($N = 10$) to +16.0 pp ($N = 30$), confirming increasing value in denser IoT deployments. (3) NOMA>OFDMA: NOMA with interference-aware pairing outperforms orthogonal-band OFDMA by +7.7 pp, confirming that SIC multiplexing benefit dominates interference cost under our pairing. Real-time feasibility: 14 ms at $N = 20$, 59 ms at $N = 50$.

Limitations

This work makes three standard simplifying assumptions whose practical implications we discuss explicitly.

Uplink/downlink asymmetry (TDD approximation). We use the uplink rate R_i as a proxy for the downlink result-return rate, justified under TDD channel reciprocity. In FDD deployments, downlink interference originates from the other UAV's transmit antennas, and the interference topology differs from the uplink case where interference comes from users of adjacent UAVs. A sensitivity check with $0.8R_i$ (20% worst-case asymmetry) shows ≤ 0.4 pp impact on our algorithm (20 seeds), confirming robustness in near-TDD scenarios. Full FDD modeling requires separate downlink SINR expressions and is left for future work.

Equal transmit power. All IoT devices transmit at $p_{\max} = 0.2\,\text{W}$ (3GPP Class 3 uplink NOMA standard [6], [31]). Joint power-pairing optimization could yield additional gains: per-pair power control ($0.8/1.2p_{\max}$) improves deadline rate by +0.9 pp (74.1% vs. 73.4%), confirming that equal-power is a sound baseline while noting that joint optimization is an open extension.

Centralized CSI and coordination overhead. Our scheme requires a central coordinator to collect global channel state information from all users to all UAVs before each scheduling slot. The CSI payload is 480 B per slot, transmitted at 10 Mbps, adding 0.38 ms overhead—negligible within the 1 s slot budget. However, this requires inter-UAV backhaul or a ground control station. Distributed or

semi-distributed approaches that operate on local CSI are a promising extension for large-scale deployments.

Optimality claims for large N. The 94.8% optimality ratio from exhaustive search at $N = 8$ is an empirical validation for small instances only. The state space grows combinatorially with N, and this ratio cannot be theoretically guaranteed for $N = 20$ or $N = 30$. We present the monotonic scaling of the swap advantage (Table VI) as empirical evidence of sustained effectiveness, not as a proof of near-optimality.

Future directions: (i) joint power-pairing [5], (ii) RSMA [32] as NOMA alternative, (iii) attention-based DRL [34] for discrete pairing.

Acknowledgment

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Data Availability

Simulation code and experiment results will be made publicly available upon acceptance.

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Appendix

Decision problem. Given a NOMA multi-UAV pairing instance with parameters $(K, N, S, \{h_{i,k}\}, t_{\max}, f_{\text{UAV}})$ and a target $W \in \mathbb{Z}^+$, does there exist a pairing assignment $\boldsymbol{\pi}$ such that at least W tasks meet the deadline?

Reduction from Independent Set. We reduce from Independent Set (IS) [9] (Problem GT20), which asks: given a graph $G = (V, E)$ and integer k, does G contain an independent set of size $\geq k$?

Construction. Given IS instance $(G = (V, E), k)$ with $|V| = n$, we construct a NOMA-Pairing instance:

1) Set $K = 2$ UAVs, and $S = k + |\mathcal{E}_{\text{ovf}}|$ subcarriers, where $\mathcal{E}_{\text{ovf}}$ is the set of overflow edges (edges of vertices with degree > 2). Set threshold $W = k$.
2) For each vertex $v_j \in V$: create a unique dummy user d_j with strong channel $h_{d_j,1} = 10^{-3}$ to UAV 1, and padding users $p_j^{(l)}$ with weak channel $h = 10^{-5.5}$. Set $f_{\text{UAV}} = f_{\text{threshold}}$ (see proof sketch in Section III) so MEC compute is non-binding and deadline satisfaction depends only on rates.
3) For each edge $(v_j, v_k) \in E$: create shared edge users $e_{jk}^{(1)}, e_{jk}^{(2)}$ with intermediate channel gains $h = 10^{-4}$.
4) Main configuration c_j for vertex v_j (weight $w = 1$): on one subcarrier, UAV 1 pairs d_j with a padding user; UAV 2 pairs the first $\min(\deg(v_j), 2)$ edge-users $\{e_{jl}^{(1)}\}$ with padding users. This configuration meets the deadline for d_j (strong channel, guaranteed positive rate).
5) Auxiliary configurations for each overflow edge $(v_j, v_k) \in \mathcal{E}_{\text{ovf}}$ (weight $w = 0$): two conflicting configurations $c_j^{(jk)}$ and $c_k^{(jk)}$ each using shared edge users $e_{jk}^{(1)}, e_{jk}^{(2)}$.

Completeness (G has IS of size $k \Rightarrow$ NOMA-Pairing achieves $W = k$). Let $\{v_{j_1}, \ldots, v_{j_k}\}$ be an independent set. Select the k main configurations $\{c_{j_l}\}$ plus auxiliary configurations $c_{j_l}^{(j_l j_m)}$ for each overflow edge. Since the vertices are independent (no shared edges), their main configurations share no edge users: the pairing is user-disjoint. Dummy users d_{j_l} achieve positive rates and meet the deadline, yielding weight k.

Soundness (NOMA-Pairing achieves $W = k \Rightarrow G$ has IS of size k). Any solution achieving weight k must select k weight-1 (main) configurations (auxiliary configurations have weight 0 by construction). Each main configuration c_j uniquely uses dummy user d_j, so k distinct vertices are identified. For any two adjacent vertices v_j, v_k sharing edge (j, k): their main configurations c_j, c_k share edge users $e_{jk}^{(1)}$ or $e_{jk}^{(2)}$, violating user-disjointness. Hence the k selected vertices form an independent set. For vertices of degree > 2 (overflow case), the overflow auxiliary configurations (weight 0) enforce disjointness via shared edge users across all incident edges: each overflow subcarrier slot is occupied by edge users $e_{jl}^{(1)}, e_{jl}^{(2)}$ for one specific edge (v_j, v_l), and user-disjointness is verified exhaustively for each overflow slot, precluding simultaneous selection of any two adjacent main configurations regardless of vertex degree.

Polynomiality. The construction uses $O(n + |E|)$ users and $O(k + |E|)$ subcarriers, both polynomial in $|G|$.

The completeness and soundness directions together establish that IS has a solution of size $\geq k$ if and only if the NOMA-Pairing instance achieves weight $\geq W = k$. Therefore, IS $\leq_p$ NOMA-Pairing. Since IS is NP-hard [9], [27], NOMA-Pairing is NP-hard. □

Note on channel values. The gains $h = 10^{-3}$, $h = 10^{-4}$, $h = 10^{-5.5}$ are realizable under the LoS path-loss model (2) at distances $\approx$112 m, 350 m, and $\approx$700 m, respectively; the 700 m distance is within the 800×800 m service area. The reduction constructs abstract instances for the general problem class and does not require these specific values; any channel gains satisfying the SIC ordering $h_{d_j,1} \gg h_{e_{jk},1} \gg h_{p,1}$ and deadline feasibility for d_j suffice.